



JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

Executive and
Professional Education



CERTIFICATE PROGRAM IN AGENTIC AI

16 WEEKS | ONLINE | LIVE MASTERCLASSES BY JHU FACULTY |
LIVE MENTORED SESSIONS BY INDUSTRY EXPERTS

THE RISE OF AGENTIC AI: FROM AUTOMATION TO AUTONOMY

Agentic AI marks a significant shift in how enterprise systems operate, moving from simple task execution to intelligent, goal-driven behavior. These systems don't just respond to instructions; they take initiative, adapt to new information, and pursue objectives with minimal oversight. Whether coordinating workflows, automating decision-making, or managing complex processes, Agentic AI brings a new level of autonomy to organizational operations.

What was once a futuristic concept is now rapidly becoming a reality. 48% of technology executives have already adopted or fully deployed Agentic AI, and half expect more than 50% of their AI systems to become autonomous within the next two years. In finance, 38% of leaders plan to adopt Agentic AI within the next 12 months. Meanwhile, 43% of retailers are piloting autonomous AI, with another 53% actively evaluating its applications.*

Unlike traditional AI, which functions within fixed boundaries, Agentic AI can reason, plan, and interact with other systems to achieve specific outcomes. This makes it a powerful force across industries, from virtual assistants that anticipate user needs to autonomous agents that accelerate research, streamline operations, and personalize services at scale.

According to the World Economic Forum, Agentic AI is projected to automate 70% of office tasks by 2030. This signals a future where AI supports business processes, collaborates in decision-making, and drives outcomes in ways that were once limited to human judgment. Agentic AI is a foundational shift in how intelligence is built and applied, designing ecosystems where machines work alongside humans, take ownership of tasks, and contribute meaningfully to organizational goals.

*Source: EY's 2025 Technology Pulse Poll, Salesforce's Connected Shoppers Report



ABOUT THE PROGRAM

The **Certificate Program in Agentic AI** is a 16-week online program designed to equip professionals with the skills to build autonomous, goal-driven AI agents capable of perceiving, reasoning, and acting independently. Combining foundational theory with hands-on Python projects, this program offers a deep dive into designing intelligent systems that can plan, learn, and collaborate to solve complex challenges.

Learners will explore agent architecture, symbolic reasoning, Belief-Desire-Intention (BDI) models, and modern techniques that incorporate Large Language Models (LLMs). The curriculum also includes reinforcement learning, multi-agent systems, and human-agent collaboration frameworks. Ethical considerations in building autonomous systems are an essential part of the learning journey.

Delivered in a flexible online format, the program allows working professionals to balance their careers and immersive learning. Learners will receive mentorship from industry experts and gain valuable insights into the world of Agentic AI.

PROGRAM HIGHLIGHTS



Duration: 16 Weeks



Format: Online (Recorded video lectures, 15 live sessions with industry experts, and monthly live masterclasses by JHU faculty)



Learning Support: Personalized assistance with a dedicated Program Manager and academic learning support via GL community, project discussion forums, and peer groups



Weekly Commitment: 8-10 hours per week



Hands on Learning: Practice with OpenAI LLMs



Recognition: Certificate of Completion and 11 Continuing Education Units (CEUs) from Johns Hopkins University, and a shareable e-Portfolio



JOHNS HOPKINS UNIVERSITY ADVANTAGE

Johns Hopkins University (JHU) is a prestigious institution consistently ranked among the top 10 universities in the U.S. and has a strong reputation in research and education. JHU is well-positioned to offer compelling differentiators for its Continuing Education Programs.

WORLD-CLASS FACULTY EXPERTISE

The courses are taught by faculty who are distinguished in academia and have industry experience leading AI practices at Fortune 500 companies. This blend of academic rigor and industry expertise offers students unique insights into the practical applications of AI. Johns Hopkins University stands at the forefront of Agentic AI innovation, with faculty who are recognized leaders in data science, neuroscience, machine learning, and national defense. Backed by a strong foundation in interdisciplinary research and global impact, the program prepares learners to build, deploy, and lead in the era of autonomous AI systems.

RESEARCH-DRIVEN CURRICULUM

The curriculum is designed based on the latest research and developments in AI, ensuring that students learn the most up-to-date and relevant information. Johns Hopkins University is the leading institution in research, consistently conducting nearly twice the R&D of any other university, and has maintained this unparalleled position for decades.

INTERDISCIPLINARY APPROACH

The courses leverage JHU's strengths across various disciplines, including computer science, engineering, and public policy. They offer students a holistic understanding of AI's impact across different sectors and enable them to think broadly about its applications.



WHO IS THIS PROGRAM FOR?

The Certificate Program in Agentic AI is designed for individuals aiming to develop and deploy AI systems that can autonomously perform tasks, make decisions, and adapt to dynamic environments.

STEM (Science, Technology, Engineering, Mathematics) Professionals: Designed for STEM professionals working in technical or tech-adjacent roles. It is ideal for learners who are familiar with programming languages or have worked with technical concepts in mathematics, data, or related fields.

Data and AI Professionals:

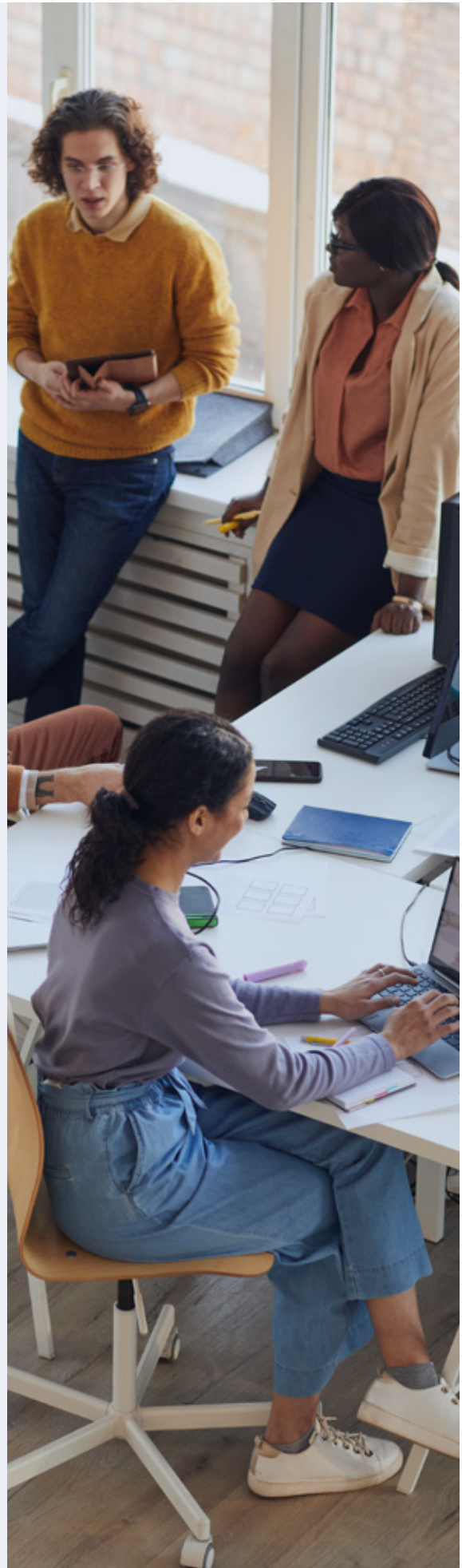
Ideal for Data Analysts, Data Scientists, AI Engineers, and ML Engineers who want to design and build intelligent agentic systems capable of autonomous decision-making and adaptability in dynamic environments.

Technical Managers and Product

Managers: Suited for professionals who want to develop a deep understanding of Agentic AI and guide their teams in transforming traditional business workflows into intelligent, autonomous agentic systems.

New Entrants to Technology:

For those who do not have prior experience with programming or technical tools, the program offers a structured Python prework module. Learners will complete an initial assessment at the end of this prework. Based on their performance, they can either move forward into the main program or take additional time to strengthen their foundations and join the next cohort when ready.



LEARNING OUTCOMES



Build Strong Conceptual Foundations

Understand Agentic AI, differentiate it from other AI paradigms, and explore planning and reasoning approaches like ReAct and Chain-of-Thought (CoT).



Design and Architect Agentic Systems

Understand and apply various agent architectures (symbolic, BDI, LLM-based) to design intelligent agents tailored for specific tasks and environments.



Implement Core Agent Capabilities

Develop agents that can perceive, reason, plan, act, and learn, utilizing Python and relevant AI libraries and frameworks.



Analyze and Evaluate Agent Behavior

Critically assess agent performance, understand the complexities of multi-agent systems and human-agent interaction, and identify risks.



Navigate Ethical Landscapes

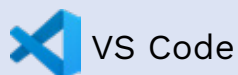
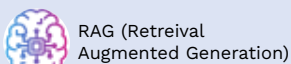
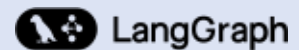
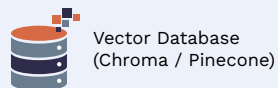
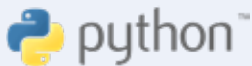
Identify, analyze, and address the ethical challenges and safety considerations inherent in developing and deploying autonomous AI systems, applying principles of responsible AI.



Apply Agentic AI to Solve Problems

Conceptualize and build functional AI agents for practical applications, demonstrating the ability to integrate diverse concepts from the program into cohesive projects.

TOOLS AND TECHNIQUES COVERED



CERTIFICATE OF COMPLETION



Note: The image is for illustrative purposes only. The actual certificate may be subject to change at the discretion of Johns Hopkins University.

PROGRAM STRUCTURE

16-WEEK PROGRAM	
PRE-WORK	PREPARATORY MODULE
3 WEEKS	PROMPT ENGINEERING FOUNDATIONS
1 WEEK	PROJECT WEEK
1 WEEK	LEARNING BREAK
3 WEEKS	INTRODUCTION TO AGENTIC AI DESIGN
1 WEEK	PLANNING & REASONING MECHANISMS
1 WEEK	PROJECT WEEK
1 WEEK	LEARNING BREAK
1 WEEK	MULTI-AGENT SYSTEMS (MAS)
2 WEEKS	ADVANCED AGENTIC AI
1 WEEK	PROJECT WEEK
1 WEEK	MASTERCLASS

PROGRAM CURRICULUM

The curriculum, designed by faculty at Johns Hopkins University and leading industry practitioners, is taught by best-in-class professors and practicing AI engineers and data professionals. The objective of the program is to equip learners with the skills needed to solve problems and deploy Agentic AI solutions across various business applications.

Note: The program also includes four Masterclasses led by JHU's best-in-class faculty on topical areas in the field. Topics and dates are subject to faculty availability.

PREPARATORY MODULE

This module will equip learners with all the tools they need for their learning journey and provide a solid foundation in Python, object oriented programming, and key data science libraries.

- Pre-Work 1: Landscape of AI, GenAI, and Agentic AI
- Pre-Work 2: Python Basics

MODULE 1: PROMPT ENGINEERING FOUNDATIONS

This module will equip learners with practical Python skills for building Agentic AI systems, introduce them to Large Language Models (LLMs), and help them understand the fundamentals of Prompt Engineering and Retrieval-Augmented Generation (RAG) for Generative AI applications.

WEEK 1: PYTHON REFRESHER FOR AGENTIC AI

This week provides a focused refresher on core Python concepts essential for building Agentic AI systems, including functions, classes, data structures, and the use of external libraries commonly applied in AI development.

- Python fundamentals
- Data structures
- Functions & classes



WEEK 2: INTRODUCTION TO LARGE LANGUAGE MODELS

This week is focused on key concepts of Generative AI, how Large Language Models (LLMs) work, what's under the hood, and why they behave the way they do. The week concludes with an exploration of Generative AI's business applications across industries and functions such as marketing, finance, healthcare, and productivity.

- LLMs
- Zero-shot prompting
- One-shot prompting
- Few-shot prompting

WEEK 3: PROMPT ENGINEERING TECHNIQUES & RAG

This week is focused on mastering essential Prompt Engineering techniques to craft effective natural language solutions. Additionally, learners will explore how Retrieval-Augmented Generation (RAG) enhances Large Language Models (LLMs) by integrating external information, improving the accuracy and relevance of AI-generated responses.

- Chain-of-thought prompting
- Vector database
- Retrieval-Augmented Generation (RAG)

WEEK 4: PROJECT 01 | SMART DATA PROCESSING AGENT

This project focuses on developing an intelligent agent to automate the processing of employee-submitted expense bills. The agent will analyze submissions and generate expense claim documents that are fully compliant with established company expense policies.

Domain: Productivity and Workflow Automation

WEEK 5: LEARNING BREAK

MODULE 2: INTRODUCTION TO AGENTIC AI DESIGN

This module will equip learners to understand the core concepts of Agentic AI, including agent properties, environment types, levels of autonomy, and goal complexity. Learners will be able to evaluate how these factors shape agent behavior across scenarios and identify key ethical, safety, and alignment challenges critical to building responsible agentic systems.

WEEK 6: INTRODUCTION TO AGENTIC AI

This week is focused on learning about Agentic AI and its key characteristics, distinguishing it from traditional AI paradigms. Learners will develop an understanding of rational agents and their role as foundational models for autonomous decision-making in intelligent systems. They will also gain insight into how agents communicate with one another using the Model Context Protocol (MCP).

- Agentic AI
- Agentic workflows
- Model Context Protocol (MCP)

WEEK 7: CORE AGENT CONCEPTS

This week is focused on the core properties of agents and the types of environments suitable for them. Learners will also compare different types of agents based on their levels of autonomy and the complexity of their goals.

- Autonomy
- Goal-directedness
- Reactivity
- Environment types (PEAS)
- Perceived agency

WEEK 8: ETHICS, SAFETY, ALIGNMENT & RESPONSIBLE AI

This week is focused on examining the AI alignment problem, exploring key ethical challenges and responsible AI principles, analyzing the unique risks posed by Agentic AI, and critically assessing agent behaviors through an ethical lens.

- Key ethical challenges (alignment, safety, responsibility)
- Responsible AI frameworks
- Agentic risks

MODULE 3: DESIGNING AND BUILDING AGENTIC SYSTEMS

This module will equip learners to understand how agents use planning and reasoning mechanisms to autonomously execute tasks and explore how multiple agents can coordinate effectively in multi agent systems to improve task efficiency, autonomy, and scalability.

WEEK 9: PLANNING & REASONING MECHANISMS

This week is focused on exploring how AI agents plan and reason by examining core paradigms like CoT and ReAct. Learners will learn to trace simple reasoning processes to solve complex tasks using Agentic AI and discover how to transform RAG into Agentic RAG.

- Planning paradigms (classical, LLM-based)
- Reasoning (CoT)
- Reasoning + acting (ReAct)
- Agentic RAG

WEEK 10: PROJECT 2 | AUTOMATED RESEARCH AGENT

This project involves creating an automated research agent designed to retrieve and synthesize information from multiple data sources, including the internet. The agent will provide comprehensive and iteratively refined answers to user queries, enhancing the depth and accuracy of information retrieval.

Domain: Intelligent Information Retrieval and Knowledge Synthesis



WEEK 11: LEARNING BREAK

WEEK 12: MULTI-AGENT SYSTEMS (MAS)

This week is focused on core concepts of Multi-Agent Systems (MAS) such as coordination and communication. Learners will analyze strategic interactions through game theory and explore the potential and challenges of integrating Large Language Models (LLMs) into MAS.

- MAS concepts (coordination, communication)
- Game theory basics
- LLM-based MAS

MODULE 4: ADVANCED AGENTIC AI

This module will equip learners to delve into the principles of Reinforcement Learning (RL) in agent systems, exploring both fundamental and advanced concepts, while comparing various learning paradigms for agent tasks and gaining insights into optimizing agent behaviors through effective learning strategies.

WEEK 13: REINFORCEMENT LEARNING IN AGENTS

This week is focused on deepening understanding of Reinforcement Learning by exploring both foundational and advanced concepts, including deep and verbal RL. Learners will evaluate how different learning paradigms support agent adaptability and lifelong learning.

- Reinforcement Learning (RL) foundations
- Advanced RL (deep RL, verbal RL)
- Lifelong learning

WEEK 14: INTERACTION & EMBODIMENT

This week is focused on exploring key concepts in human-agent interaction, such as trust and common ground. Learners will understand the role of simulation in embodied AI and examine the challenges in effective communication and deployment of embodied agents.

- Human-agent interaction (HAI) (trust, common ground)
- Embodied AI (simulation, VLA models)

WEEK 15: PROJECT 3 | CUSTOMER SUPPORT CHATBOT WITH KNOWLEDGE BASE INTEGRATION

This project entails the development of a customer support chatbot engineered to address frequently asked questions (FAQs) concerning a specific product or service. The agent will leverage a predefined knowledge base, employ advanced knowledge retrieval techniques, engage in natural language interactions, and integrate a Large Language Model (LLM) to enhance communication clarity and user experience.

Domain: Customer Support & Automation

WEEK 16: THE FUTURE OF AGENTIC AI

This week is focused on helping learners deepen their understanding of Agentic AI concepts, applications, and strategic implications for their organizations. The sessions will explore the current limitations of Agentic AI, including brittleness, scalability, and memory. They will analyze its role in the pursuit of Artificial General Intelligence (AGI), identify key research challenges in robust reasoning, continual learning, and autonomous tool use, and evaluate the societal and economic impacts of future agentic technologies.

PROGRAM FACULTY AND INDUSTRY MENTORS

FACULTY



Dr. Shelby Wilson is a distinguished Senior Data Scientist at the Johns Hopkins University Applied Physics Laboratory (JHU/APL) and holds a faculty position at Johns Hopkins University Engineering for Professionals. With a profound expertise in applied mathematics, mathematical biology, and data science, her work encompasses computational epidemiology, dynamical systems, and machine learning to model complex biological phenomena. Her leadership extends to her role as the Associate Director for Education, Training, Outreach, and Participation for the NSF Center for the Analysis and Prediction of Pandemic Expansion (APPEX), where she contributes to national efforts in public health. In recognition of her significant scientific contributions to the field and her commitment to diversity, Dr. Wilson was inducted as a fellow of the Association for Women in Mathematics in 2022.

Dr. Wilson earned a dual Bachelor of Science degree in Mathematics and Computer Science from Spelman College. She then earned her Ph.D. in Applied Mathematics from the University of Maryland, College Park, where her research focused on mathematical models of immune regulation and cancer vaccines. Her career includes a postdoctoral fellowship at the French National Institute for Research in Digital Science and Technology INRIA, in Grenoble, France, and faculty positions at both Morehouse College and the University of Maryland before joining JHU/APL. She has over a decade of experience in developing and applying mathematical and computational approaches to solve critical research problems.



Dr. Ian McCulloh holds joint faculty appointments in Public Health and Engineering at Johns Hopkins University. His current research focuses on strategic influence in online networks and the cognitive dimension of the information environment. The author of three books and over 46 peer-reviewed papers, primarily in social network analysis, Dr. McCulloh has also served as Chief Data Scientist and Managing Director of AI at Accenture Federal Services. He retired from the U.S. Army after 20 years of service in special operations and C-IED forensics and targeting. He holds a PhD and three master's degrees.



Dr. William Gray-Roncal, is a Principal Research Scientist at the Johns Hopkins University Applied Physics Laboratory, where he leads interdisciplinary research in areas such as neuroscience, artificial intelligence, machine learning, and healthcare. With over 20 years of professional experience as an electrical engineer across diverse domains including space, submarines, and biomedicine, his current projects focus on research to estimate and understand brain networks at the synaptic scale, and to explore innovative approaches for rapid scene analysis and decision making. His work frequently integrates learning and education research into his science and engineering projects. Through efforts including the College Prep and CIRCUIT programs, he has developed models that identify and support top STEM talent to become leaders in their fields, while simultaneously achieving innovative solutions with student research collaborations. Over the last two decades, he has mentored nearly 500 students.

His research has made significant contributions to the field of connectomics, with expertise spanning diverse areas including computer vision, signal and image processing, machine learning, network analysis, artificial intelligence, and systems engineering. He recently published a paper as part of a Nature special collection highlighting a large-scale research effort to map the structural and functional connectivity of a cubic millimeter of mouse visual cortex at nanoscale resolution. This work has implications for neuroscience understanding, health applications, and next-generation AI algorithms. Dr. Gray-Roncal holds a Ph.D. in Computer Science from Johns Hopkins University, an M.S. in Electrical Engineering from the University of Southern California, and a B.S. in Electrical Engineering from Vanderbilt University. He has academic appointments in Computer Science, Johns Hopkins Engineering Executive and Professional Education within the Whiting School of Engineering, as well as an appointment in the JHU School of Education. Will lives in Columbia, Maryland, with his wife, son, and two dogs, Habiba and Myelin. He enjoys music, backpacking, and traveling.



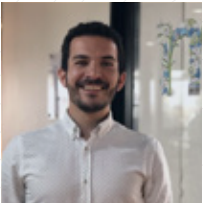
Dr. Iain Cruickshank holds faculty appointments in Computer Science at Johns Hopkins University and Carnegie Mellon University. He is a practicing data scientist and Artificial Intelligence (AI) engineer for the U.S. Army. During his tenure with the U.S. Army, Iain has been a data scientist at the Army's Artificial Intelligence Integration Center, a senior research scientist at the Army Cyber Institute, and has held various leadership and technical positions within the U.S. Army. His research is focused on novel, machine learning-based techniques for understanding online information environments and platforms, and on the test, evaluation, and acquisition of machine-learning-enabled systems. Techniques from his research have been used by industry and academic research groups to analyze online misinformation spread and state-sponsored disinformation campaigns. He has published over 40 peer-reviewed academic papers. Dr. Cruickshank received a Ph.D. in Societal Computing from Carnegie Mellon University, a Master of Science in Operations Research from the University of Edinburgh, and a Bachelor of Science in Mathematics from the United States Military Academy at West Point. His honors include a National Science Foundation Graduate Research Fellowship and a Rotary Ambassadorial Scholar.

INDUSTRY MENTORS



BHASKARJIT SARMAH

Head of AI Research
at Domyen



KALLE BYLIN

Data Engineer at
Workday



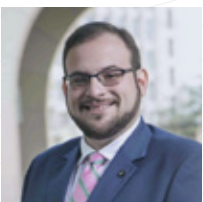
TANYA GLOZMAN

Applied Science - AI/ML
at Apple



BRIDGET HUANG

GenAI/ML Engineer at Amazon
Web Services (AWS)



VINICIO DE SOLA

Staff Data Scientist
at Newmark

Note: This is an indicative list and is subject to change based on the availability of faculty and mentors.



ADMISSION PROCESS



The admission process is conducted on a rolling basis and will be closed once the requisite number of candidates are enrolled.



APPLY

Fill out an online application form



REVIEW

Eligible applications will be reviewed by a panel from Great Learning



JOIN PROGRAM

An offer letter will be sent to the selected candidates

PROGRAM FEE

\$3,000

Please contact the Program Advisor from Great Learning for more information on offers, payment plans, and eligibility for financial assistance.

PROGRAM PARTNER



Great Learning is a leading global ed-tech company for professional and higher education. It offers comprehensive, industry-relevant programs across various cutting-edge Technology, Data, and Business domains. These programs are developed in collaboration with the world's foremost academic institutions, such as Johns Hopkins University, MIT Professional Education, The University of Texas at Austin, Northwestern School of Professional Studies, Deakin University, Great Lakes Institute of Management, and more. They are constantly reimagined and revamped to address the dynamic needs of the industry. Offered in blended, classroom, and purely online modes, these programs are delivered with the help of expert mentors and highly qualified faculty. Great Learning is on a mission to enable transformative learning and career success in the digital economy, which has impacted 14 million+ learners from over 170 countries.



14 Mn+
Learners



170+
Countries



8000+
Industry Experts



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APPLY NOW
